

Multispectral Water Body Segmentation Report

1 Objective

The primary objective of this project is to develop a reliable deep learning system capable of segmenting water bodies using multispectral satellite imagery. Accurate water boundary detection is essential for monitoring water resources, managing floods, and environmental conservation. Multispectral data provides a significant advantage over standard optical images, as it captures wavelengths outside the visible spectrum, allowing for a more distinct separation between water, vegetation, and land.

2 Tasks and Requirements

The project fulfills the following core requirements:

- **Dataset Exploration:** The dataset consists of 306 total samples, divided into 244 training images and 62 validation images. Each image contains 12 spectral bands with a spatial resolution of 128 by 128 pixels. The target labels are provided as binary masks, where a pixel value of 1 represents water and 0 represents the background.
- **Band Visualization:** The 12-channel input includes standard visible light (Red, Green, Blue) alongside specialized sensor data such as Near-Infrared (NIR) and Shortwave Infrared (SWIR). These additional bands are critical because water heavily absorbs infrared light, causing it to appear distinctly dark compared to highly reflective land and vegetation. Figure 1 displays the individual layers of the dataset.
- **Preprocessing:** The data pipeline individually normalizes each of the 12 spectral bands. This ensures that features with naturally higher intensity values do not overpower the network during training. The final output is formatted as a standardized tensor with the shape (12, 128, 128).
- **Model Architecture:** A U-Net model was implemented from scratch to map the 12-channel input to a single-channel binary output. The U-Net architecture consists of two main sections: an encoder that reduces the image size to learn broad feature patterns, and a decoder that scales the image back to its original dimensions. Skip connections link the encoder directly to the decoder, transferring exact pixel locations. This allows the model to understand both the context of the image and the precise boundaries of the water bodies.
- **Training and Evaluation:** The model's performance was evaluated using Intersection over Union (IoU), precision, recall, and F1-score. These metrics ensure

that the model is penalized equally for missing water bodies (false negatives) and mislabeling land as water (false positives).

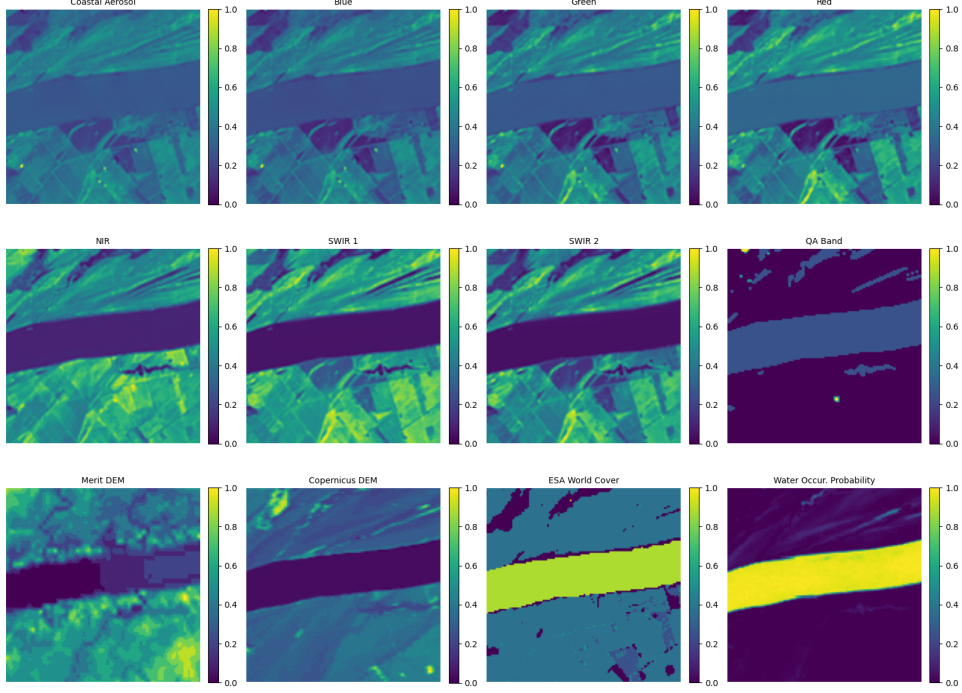


Figure 1: Visualization of the 12 spectral bands from the satellite dataset.

3 Training and Optimization

3.1 Layer Normalization Issue

During early training runs, the Validation IoU metric failed to improve, immediately dropping to exactly zero and remaining frozen. This issue was traced back to the Batch Normalization layers. Image segmentation models often require restricted batch sizes due to memory constraints. Batch Normalization relies on computing the average and variance across these batches; when batches are small, these calculations fluctuate heavily.

During the validation phase, the model stops updating and defaults to using the accumulated running averages. Because the small-batch averages were unstable and unrepresentative of the overall data, the network’s activations shifted drastically, causing the model to fail on validation data. To resolve this, Batch Normalization was replaced with Group Normalization. Group Normalization divides the 12 channels into groups and normalizes them for each individual image, making the process completely independent of the batch size. This restored the model’s ability to learn and update the validation metric properly. This frozen metric behavior is documented in Figure 2.

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Dataset summary: 306 total samples | 244 Train | 62 Val
Starting training on device: cuda
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Epoch 01/100 | Train Loss: 0.6382 | Train IoU: 0.2117 || Val Loss: 0.6703 | Val IoU: 0.0000 -> Saved new best model (Val IoU: 0.0000)
Epoch 02/100 | Train Loss: 0.5685 | Train IoU: 0.3175 || Val Loss: 0.6245 | Val IoU: 0.0000
Epoch 03/100 | Train Loss: 0.5341 | Train IoU: 0.3225 || Val Loss: 0.6113 | Val IoU: 0.0000
Epoch 04/100 | Train Loss: 0.5109 | Train IoU: 0.3350 || Val Loss: 0.6020 | Val IoU: 0.0000
Epoch 05/100 | Train Loss: 0.5168 | Train IoU: 0.2824 || Val Loss: 0.6023 | Val IoU: 0.0000
Epoch 06/100 | Train Loss: 0.4735 | Train IoU: 0.3907 || Val Loss: 0.6014 | Val IoU: 0.0000
Epoch 07/100 | Train Loss: 0.4962 | Train IoU: 0.2936 || Val Loss: 0.6024 | Val IoU: 0.0000
Epoch 08/100 | Train Loss: 0.5003 | Train IoU: 0.2589 || Val Loss: 0.6016 | Val IoU: 0.0000
Epoch 09/100 | Train Loss: 0.4526 | Train IoU: 0.3807 || Val Loss: 0.6017 | Val IoU: 0.0000
Epoch 10/100 | Train Loss: 0.4544 | Train IoU: 0.3685 || Val Loss: 0.6032 | Val IoU: 0.0000
Epoch 11/100 | Train Loss: 0.4705 | Train IoU: 0.2875 || Val Loss: 0.6006 | Val IoU: 0.0000
Epoch 12/100 | Train Loss: 0.4200 | Train IoU: 0.4102 || Val Loss: 0.6028 | Val IoU: 0.0000
Epoch 13/100 | Train Loss: 0.4729 | Train IoU: 0.2755 || Val Loss: 0.6133 | Val IoU: 0.0000
Epoch 14/100 | Train Loss: 0.4267 | Train IoU: 0.3806 || Val Loss: 0.6080 | Val IoU: 0.0000
Epoch 15/100 | Train Loss: 0.4580 | Train IoU: 0.2908 || Val Loss: 0.6151 | Val IoU: 0.0000
Epoch 16/100 | Train Loss: 0.4496 | Train IoU: 0.3291 || Val Loss: 0.6190 | Val IoU: 0.0000
Epoch 17/100 | Train Loss: 0.4267 | Train IoU: 0.3704 || Val Loss: 0.6014 | Val IoU: 0.0000
Epoch 18/100 | Train Loss: 0.4212 | Train IoU: 0.3699 || Val Loss: 0.6046 | Val IoU: 0.0000
Epoch 19/100 | Train Loss: 0.4474 | Train IoU: 0.3065 || Val Loss: 0.6011 | Val IoU: 0.0000
Epoch 20/100 | Train Loss: 0.3982 | Train IoU: 0.4206 || Val Loss: 0.6426 | Val IoU: 0.0000
Epoch 21/100 | Train Loss: 0.4071 | Train IoU: 0.3905 || Val Loss: 0.6100 | Val IoU: 0.0000
Epoch 22/100 | Train Loss: 0.4376 | Train IoU: 0.2997 || Val Loss: 0.5998 | Val IoU: 0.0000

```

Figure 2: Training log showing the Validation IoU metric frozen at 0.0000 due to Batch Normalization limitations.

3.2 Dataset Shuffling

An initial training attempt revealed a severe overfitting problem: the training loss consistently decreased while the validation loss diverged and increased. This occurred because the dataset was processed in the exact same static order during every epoch. Consequently, the model began memorizing the sequence of the images rather than learning the generalized visual features of water.

By implementing dataset shuffling at the beginning of each training epoch, the input order was randomized. This modification broke the predictable sequence and forced the model to update its parameters based on a diverse, random distribution of images, which successfully aligned the training and validation loss trajectories. The divergence caused by the lack of shuffling is illustrated in Figure 3.

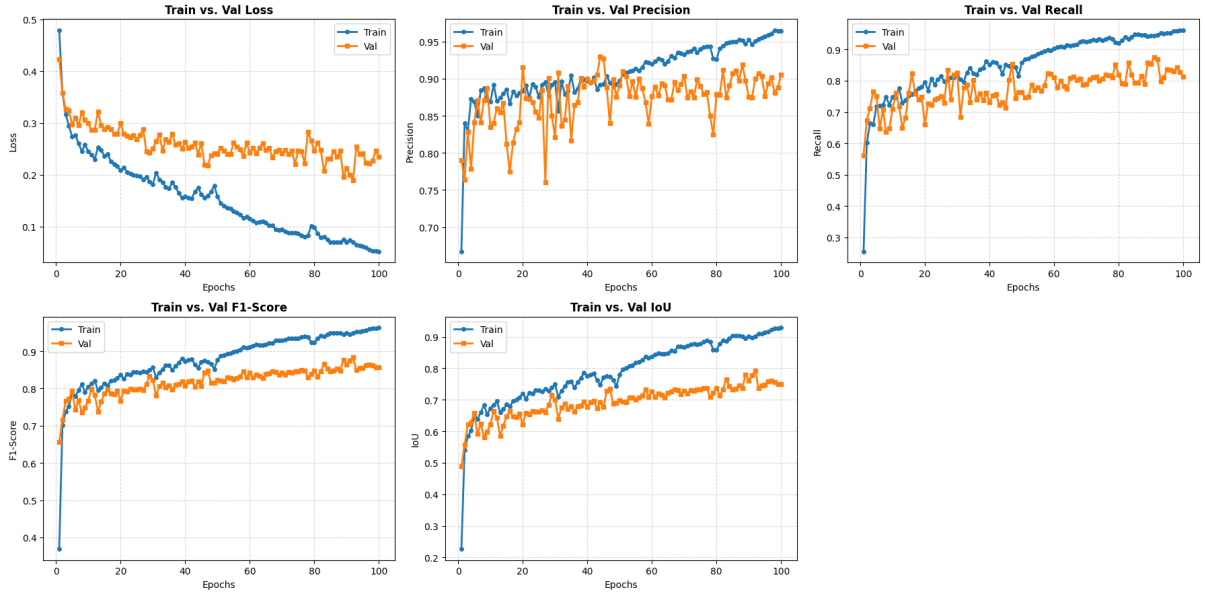


Figure 3: Metrics showing diverging training and validation loss when the dataset was not shuffled.

3.3 Model Evaluation

The model was tested across 150 epochs using two different learning rates to evaluate convergence and stability. As shown in Figure 4, a learning rate of 10^{-4} resulted in a faster initial optimization but caused the precision and recall metrics to fluctuate heavily during the early epochs. Conversely, Figure 5 demonstrates that a learning rate of 10^{-5} provided a smoother and more stable training curve, with the training and validation metrics tracking each other closely. Both runs confirm that the final model trains without errors and successfully captures the water class with a high IoU.

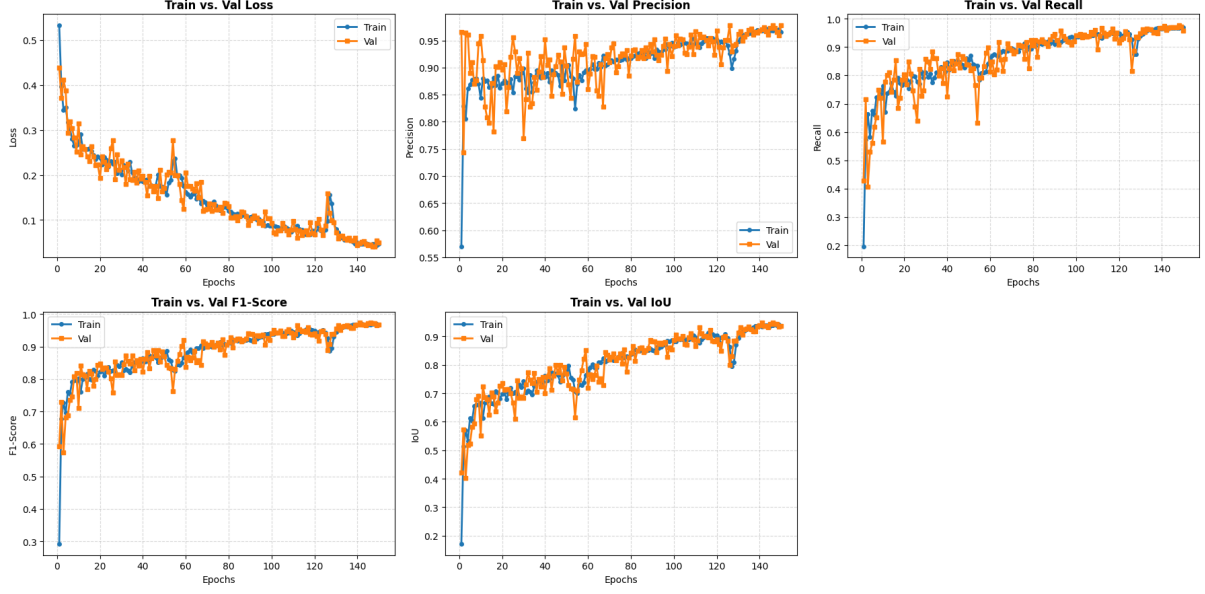


Figure 4: Training and validation metrics using a learning rate of 10^{-4} .

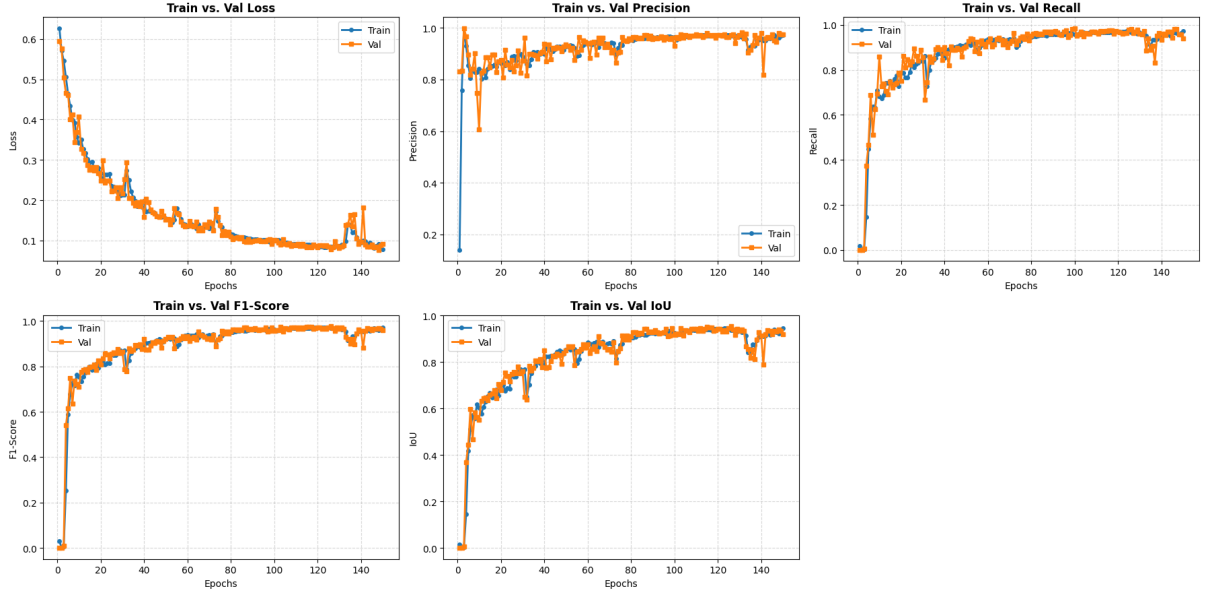


Figure 5: Training and validation metrics using a learning rate of 10^{-5} .

4 Conclusion

The completed pipeline successfully processes 12-channel multispectral images and trains a customized U-Net model. By diagnosing the batch statistics failure and switching to Group Normalization, alongside enforcing dataset shuffling, the training process was fully stabilized. The final evaluation metrics confirm that the network can reliably and accurately segment water bodies from complex satellite imagery.